

Assignment - javascript

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Type: DESCRIPTIVE

DESCRIPTIVE QUESTIONS:

1. Explain the core JavaScript concepts of scope, hoisting, and closures. Provide a detailed code example for each, illustrating how they function and what potential implications they have for variable accessibility and function behavior in complex applications.
2. Discuss the impact of modern ECMAScript features (ES6+) on contemporary JavaScript development. Select three distinct features (e.g., arrow functions, destructuring, async/await, modules) and thoroughly explain their syntax, practical use cases, and how they enhance code readability, maintainability, or performance compared to older JavaScript paradigms.
3. Describe a significant real-world application where JavaScript is a critical component, either on the client-side, server-side (Node.js), or both. Detail the architectural choices made regarding JavaScript frameworks/libraries, how data flows through the application, and specific challenges overcome by leveraging JavaScript's capabilities.
4. You are tasked with debugging a large, complex JavaScript application experiencing intermittent issues. Outline a comprehensive strategy for identifying, diagnosing, and resolving performance bottlenecks and unexpected runtime errors. What tools, methodologies, and best practices would you employ throughout this process?
5. Elaborate on the role and importance of JavaScript build tools (e.g., Webpack, Vite, Parcel) and package managers (e.g., npm, Yarn) in a modern front-end or full-stack development workflow. Explain how these tools integrate, streamline the development process, manage dependencies, and optimize application deployment.

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